

A* Pathfinding Algorithm

This project aims to create a flexible implementation of the A* pathfinding algorithm based upon an improved version of the Graph data structure from Lab 13.

A* is a common pathfinding algorithm which is used to determine the most optimal path between two graph vertices while minimizing the time spent visiting unnecessary nodes. This is achieved by using a heuristic function to estimate the *remaining* cost to reach the goal after traversing to a previously unexplored node. The algorithm greedily selects the next node with the lowest current + estimated cost.

The ultimate goal of the project is to create an API for A* in such a way that both the underlying Graph datatype and the heuristic function can be easily modified or entirely replaced to fit the needs of a specific implementation.

This implementation of A* will utilize a priority queue to track the estimated total cost of known but unvisited nodes to visit next. A hash table will also be used to track the known cost of previously-visited nodes, and the current shortest path to those nodes.

The user interface requirement will utilize a modified version of the UI from Lab 13 as a proof-of-concept implementation. The UI will:

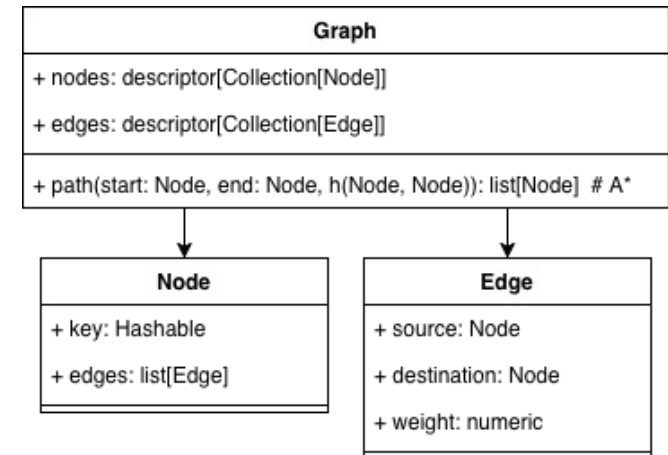
- Allow users to view a printed image of the graph (utilizing a library such as matplotlib)
- Allow adding new nodes/edges
- Allow users to request the shortest path between two nodes
- Visualize the stepwise execution of A* and its result on the graph

Test Plan:

1. Create a test graph with ~20 nodes
2. Run Dijkstra's algorithm on a node to determine the optimal path length to each other node
3. Run A* from that node to all other nodes, ensure path length matches Dijkstra.

Timeline:

- 5/5: Graph data structure complete (part of Lab 13)
- 5/7: A* implemented
- 5/10: Modifications to Lab 13 UI underway
- 5/12: Modifications to UI complete, testing underway
- 5/14: Presentation
- 5/19: Code Turned In



Impact & Limitations:

With no additional heuristic beyond the weight of edges between nodes, A* is effectively a version of Dijkstra's algorithm which terminates once the target node is found. However, the primary power of A* is the ability to customize the heuristic function based on the properties of the graph.

For instance, if the graph represents a road network where each node is an intersection, the geographic linear distance between each node and the target can be incorporated into the heuristic function, "guiding" A* to prefer steps which lower that distance over those that increase it.